

BUBBLE SORT:

Bubble Sort is a simple sorting algorithm that repeatedly compares adjacent elements and swaps them if they are in the wrong order. This process continues until the entire array is sorted.

ALGORITHM:

1. Start from the first element of the array.
2. Compare two adjacent elements.
3. If the first element is greater than the second, swap them.
4. Continue comparing and swapping adjacent elements until the end of the array.
5. Repeat the process until no more swaps are needed.

Time Complexity:

Case	Time Complexity
Best Case (Already Sorted)	$O(n)$
Average Case	$O(n^2)$
Worst Case (Reverse Sorted)	$O(n^2)$

Space Complexity:

Space Complexity = $O(1)$

OUTPUT:

Enter the number of elements: 5

Enter 5 elements:

7

5

8

3

1

Sorted array: 1 3 5 7 8

...Program finished with exit code 0

Press ENTER to exit console.